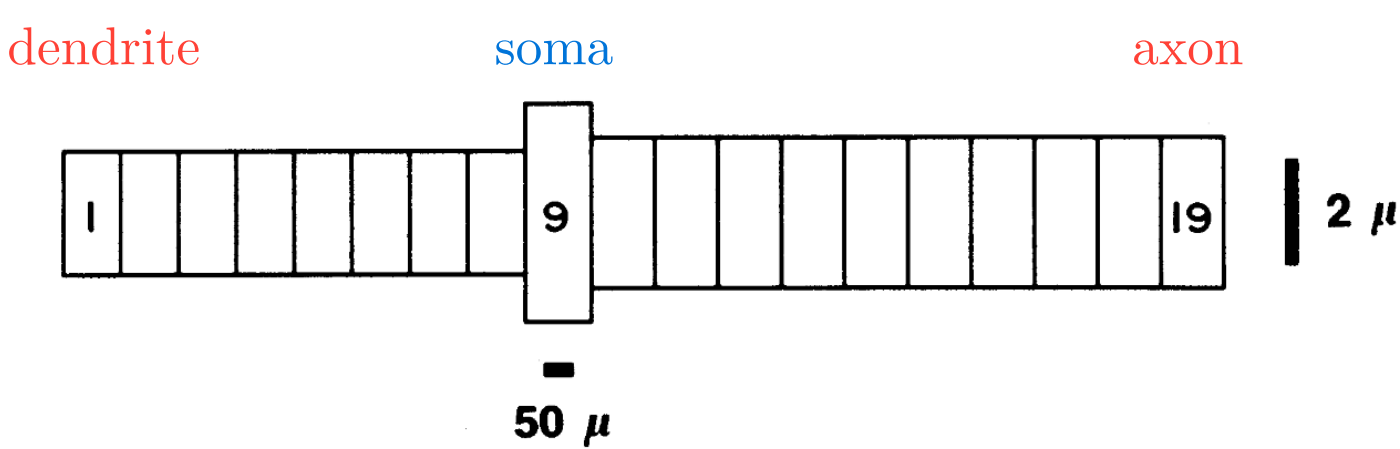


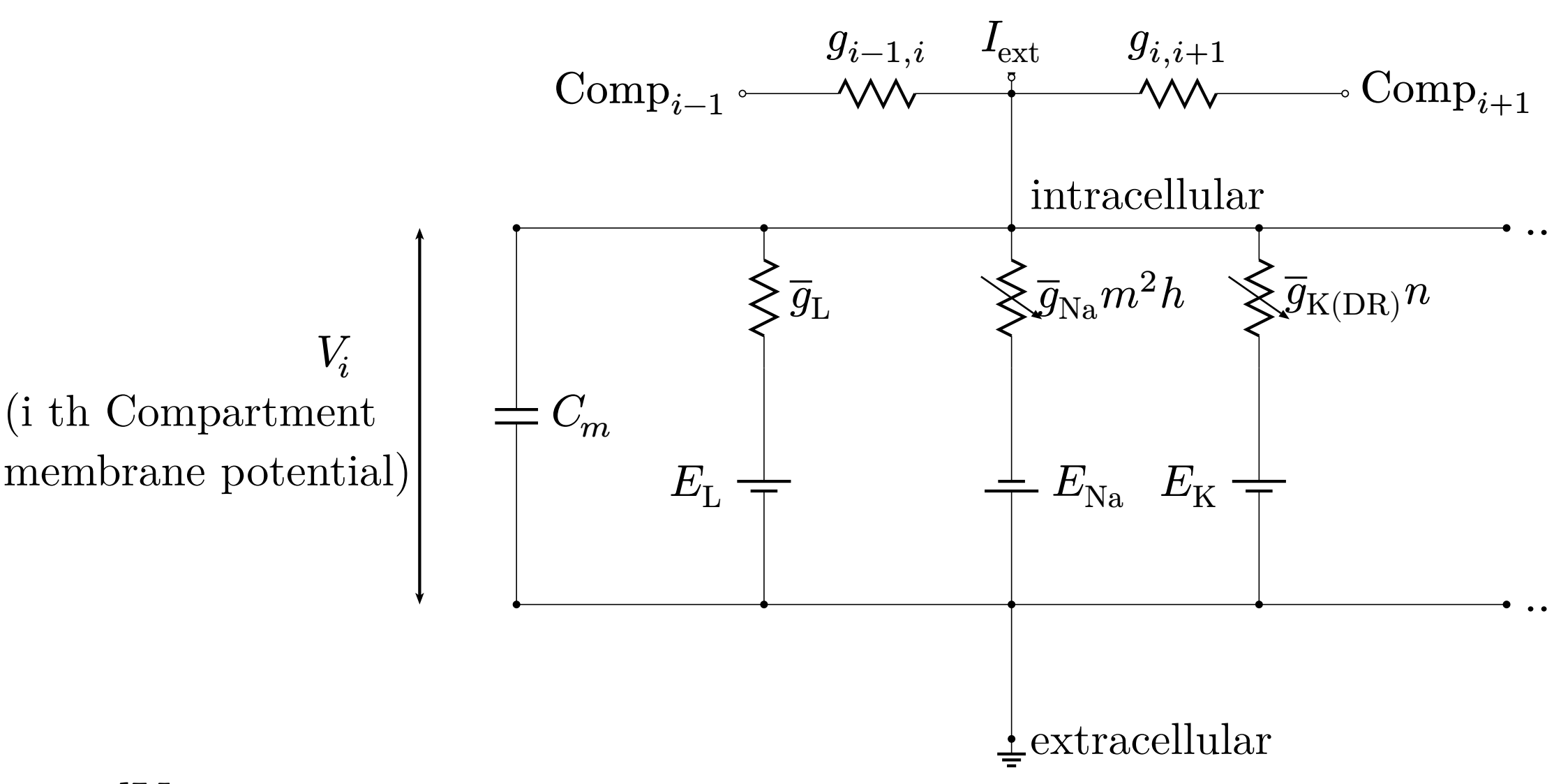
Introduction



Guinea pig CA3 pyramidal neuron[1]



Modelled as **19 Compartments(comps)** [2]



(i th Compartment membrane potential)

$$C_m \frac{dV}{dt} = -g_{\text{leak}}(V - V_{\text{leak}}) - \bar{g}_{\text{Na}} m^2 h (V - V_{\text{Na}}) - \bar{g}_{\text{K}} n (V - V_{\text{K}}) + I_{\text{ext}}$$

Conductances have **gate variables**, ODEs with rate functions α, β :

$$I_{\text{Na}} = \bar{g}_{\text{Na}} m^2 h (V - E_{\text{Na}})$$
$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$
$$\alpha_m(V) = 0.32(13.1 - u)/(\exp((13.1 - u)/4) - 1)$$
$$\beta_m(V) = 0.28(u - 40.1)/(\exp((u - 40.1)/5) - 1)$$

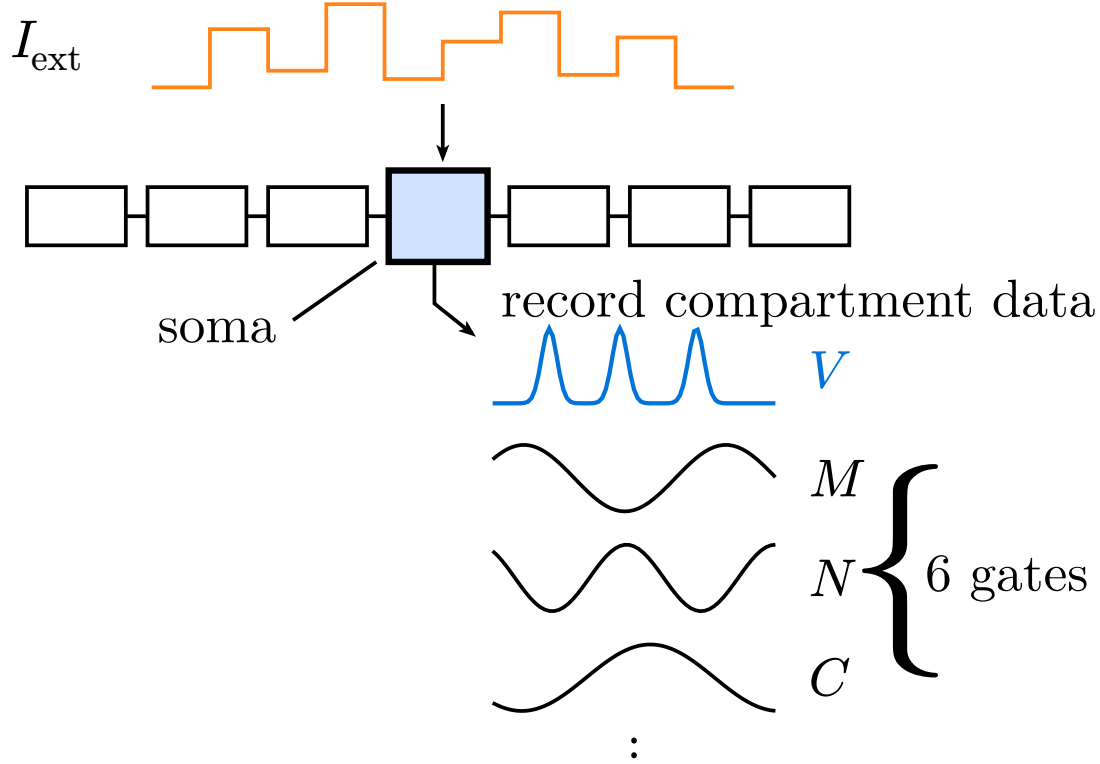
→ **11 states per compartment**
→ **209** for 19 comps;
At large scale network simulation, the gates become a **memory bottleneck**.

Goal

Development of a multi-compartment neuron’s surrogate model capable of reproducing the membrane potential response with fewer gate variables.

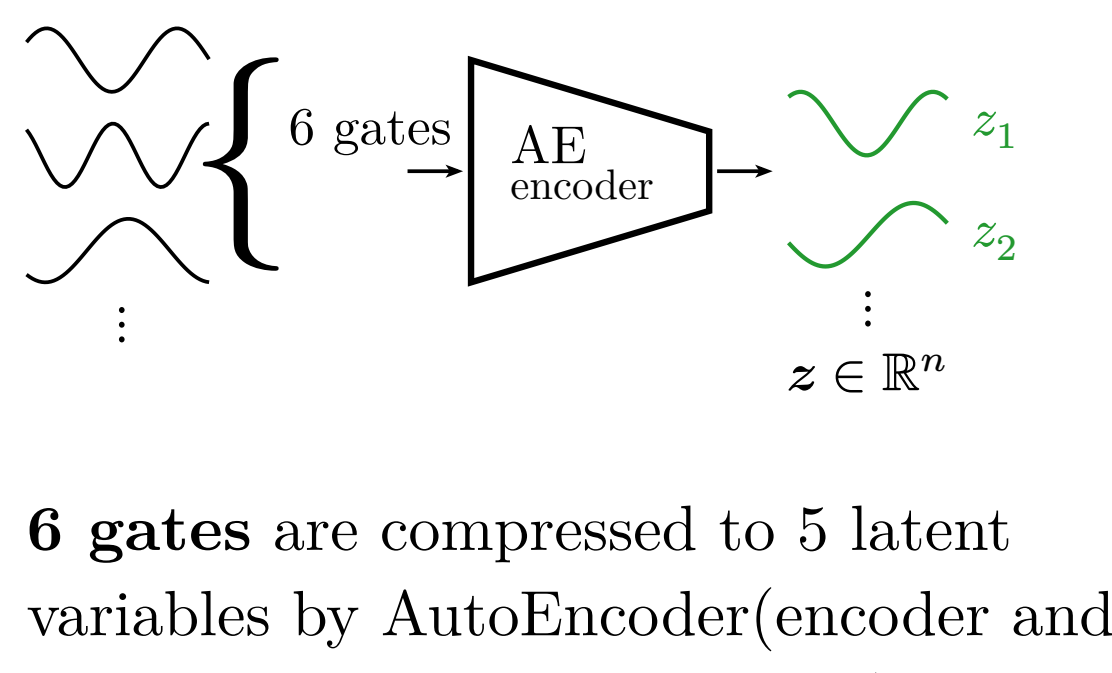
Methods

① Sample the training data



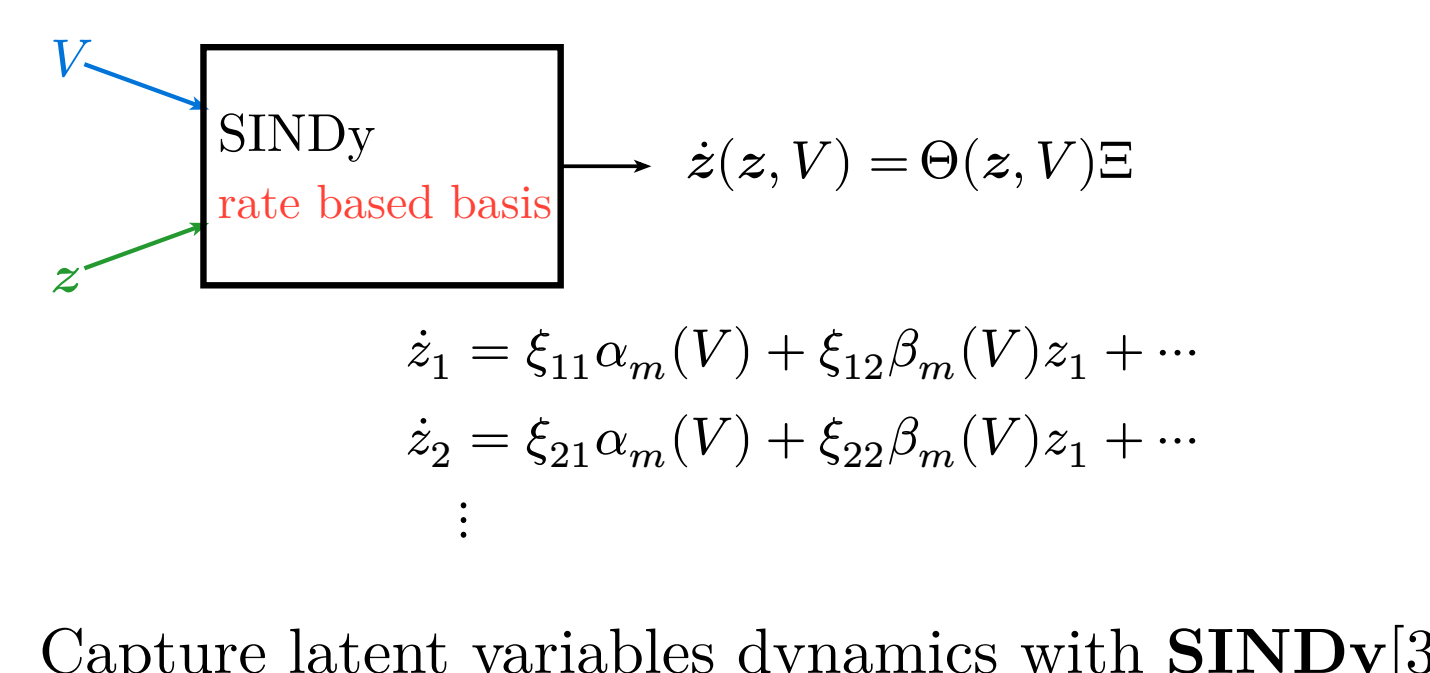
Inject a **random pulse train** at the soma; record V and **6 gates**.

② Compress gates



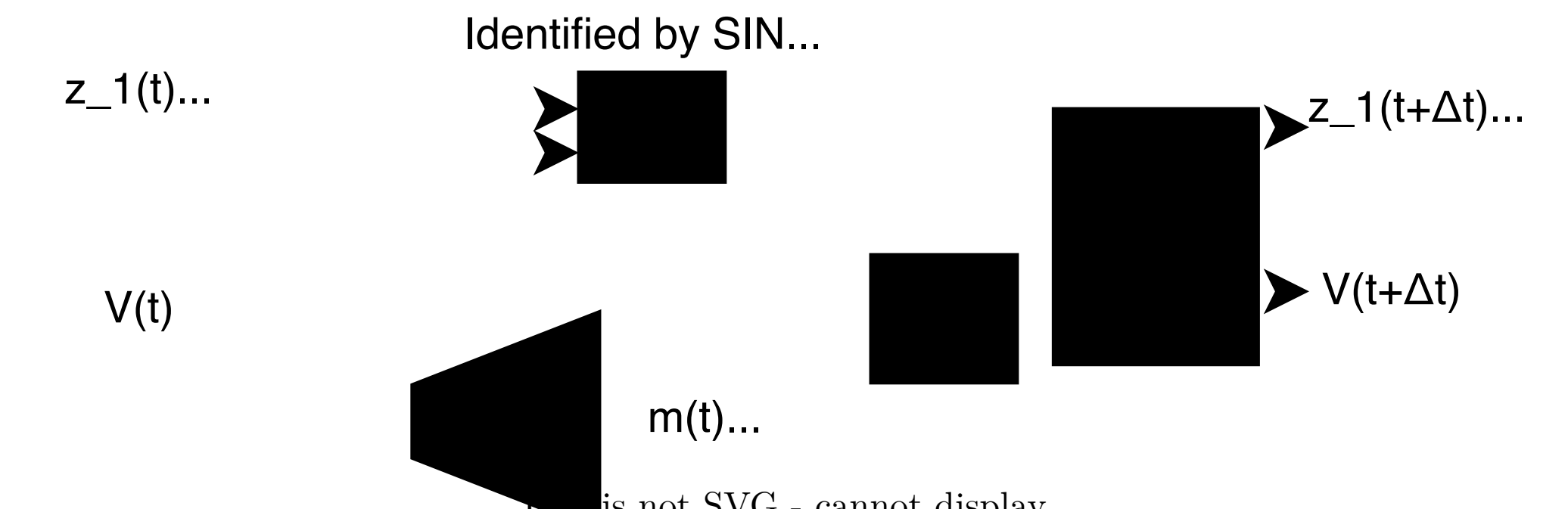
6 gates are compressed to 5 latent variables by AutoEncoder(encoder and decoder have one hidden layer).

③ Identify the latent variables of ODEs



Capture latent variables dynamics with **SINDy**[3]. SINDy fits coefficients Ξ

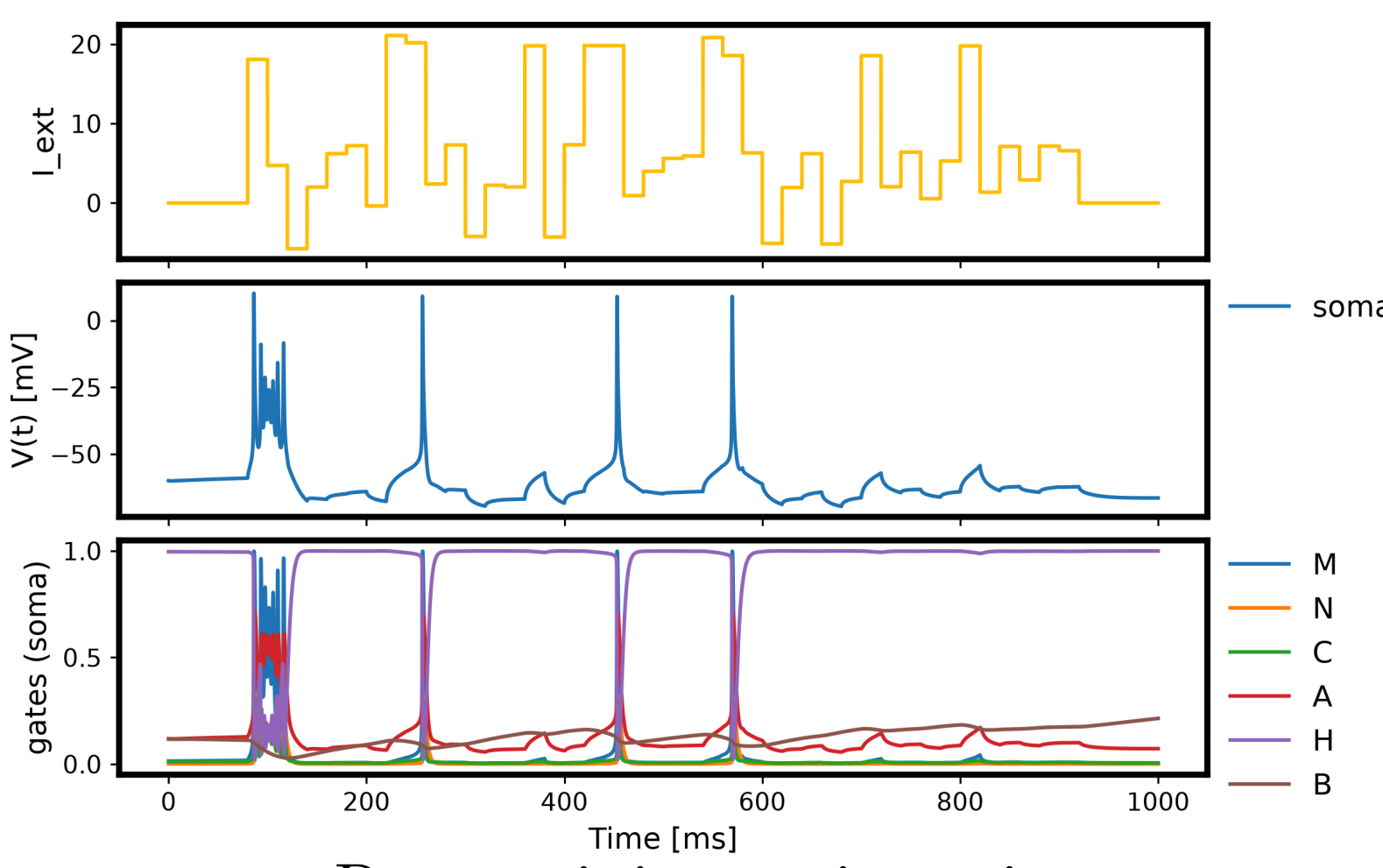
④ How to apply the surrogate model



The derivative of V is computed with the **decoded latent variables**. Across time steps, only the latent variables and the membrane potential need to be preserved.

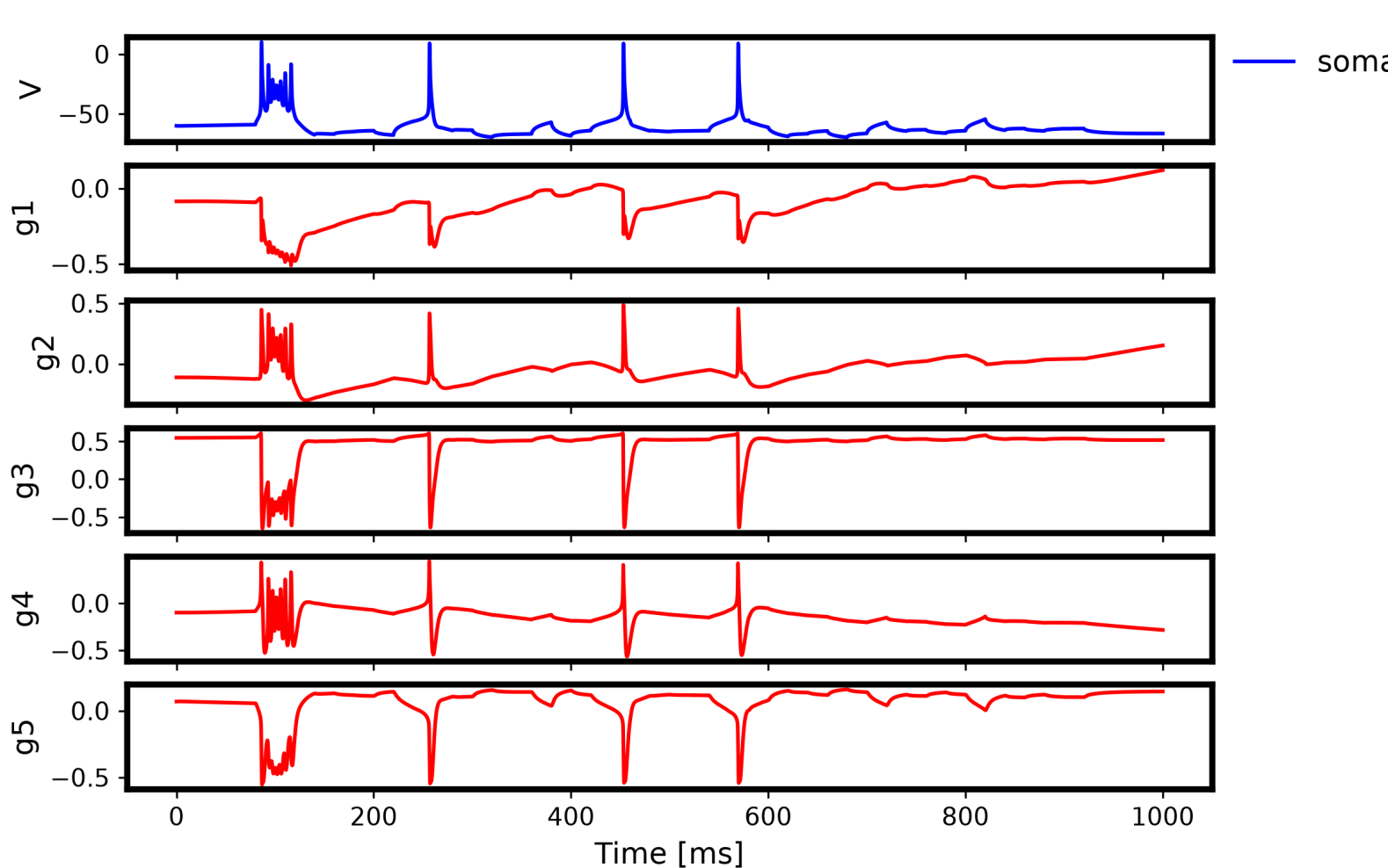
Results and Discussion

Training Data to capture gate dynamics



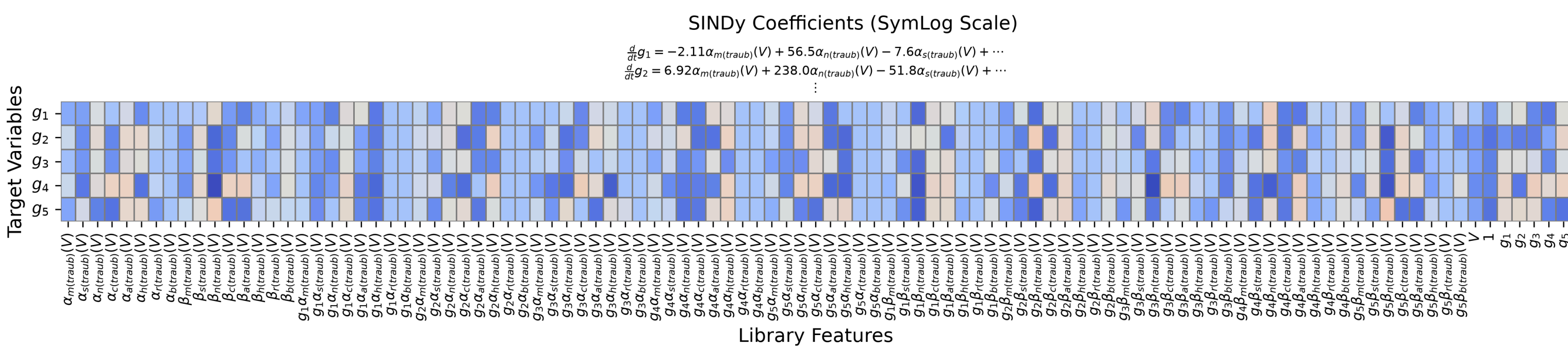
Raw training trajectories.

↓ compress



Latent variables and membrane potential used to fit SINDy.

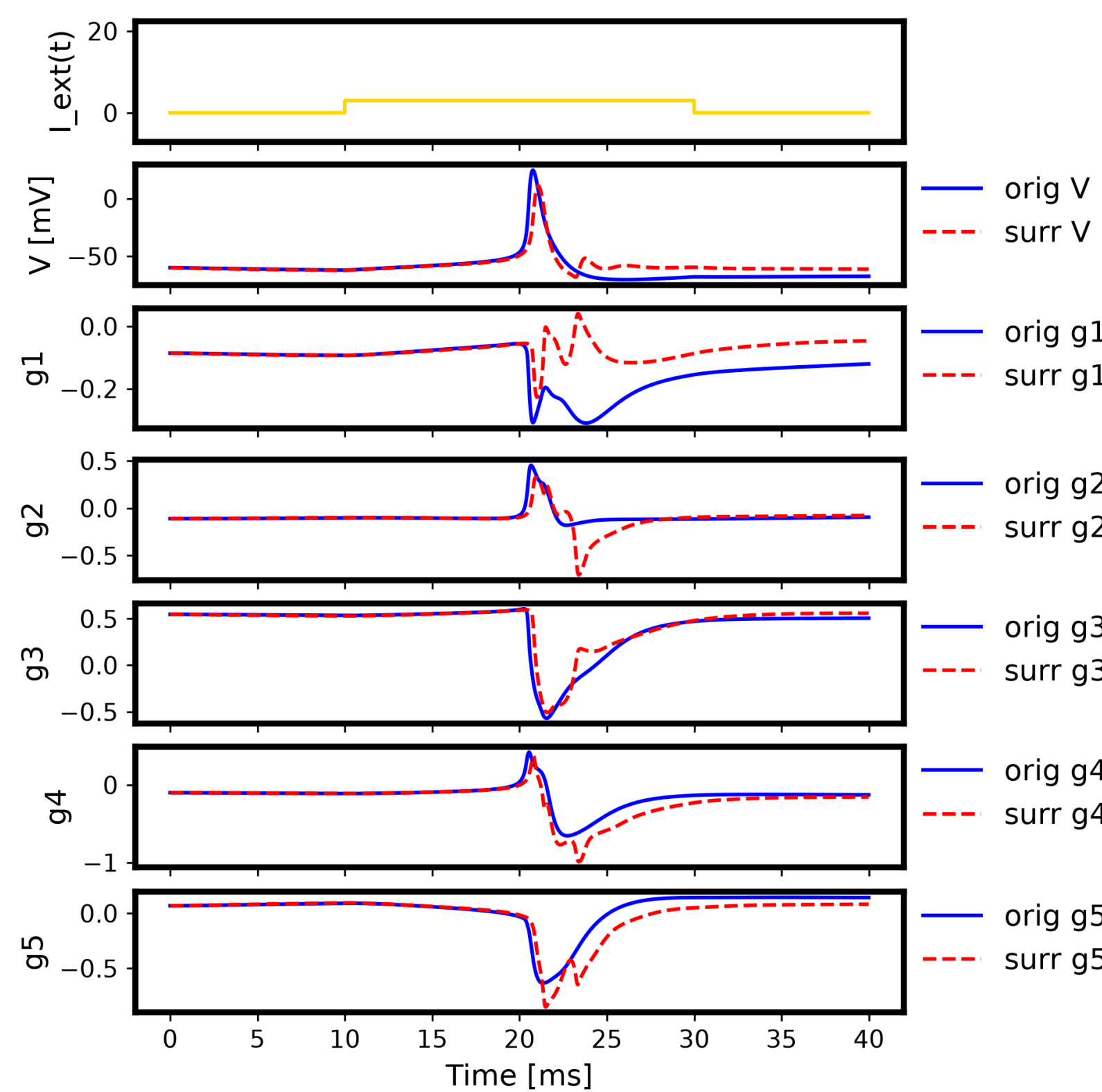
Identified latent equations



SINDy Coefficients (SymLog Scale)

ξ over the physics-informed library.

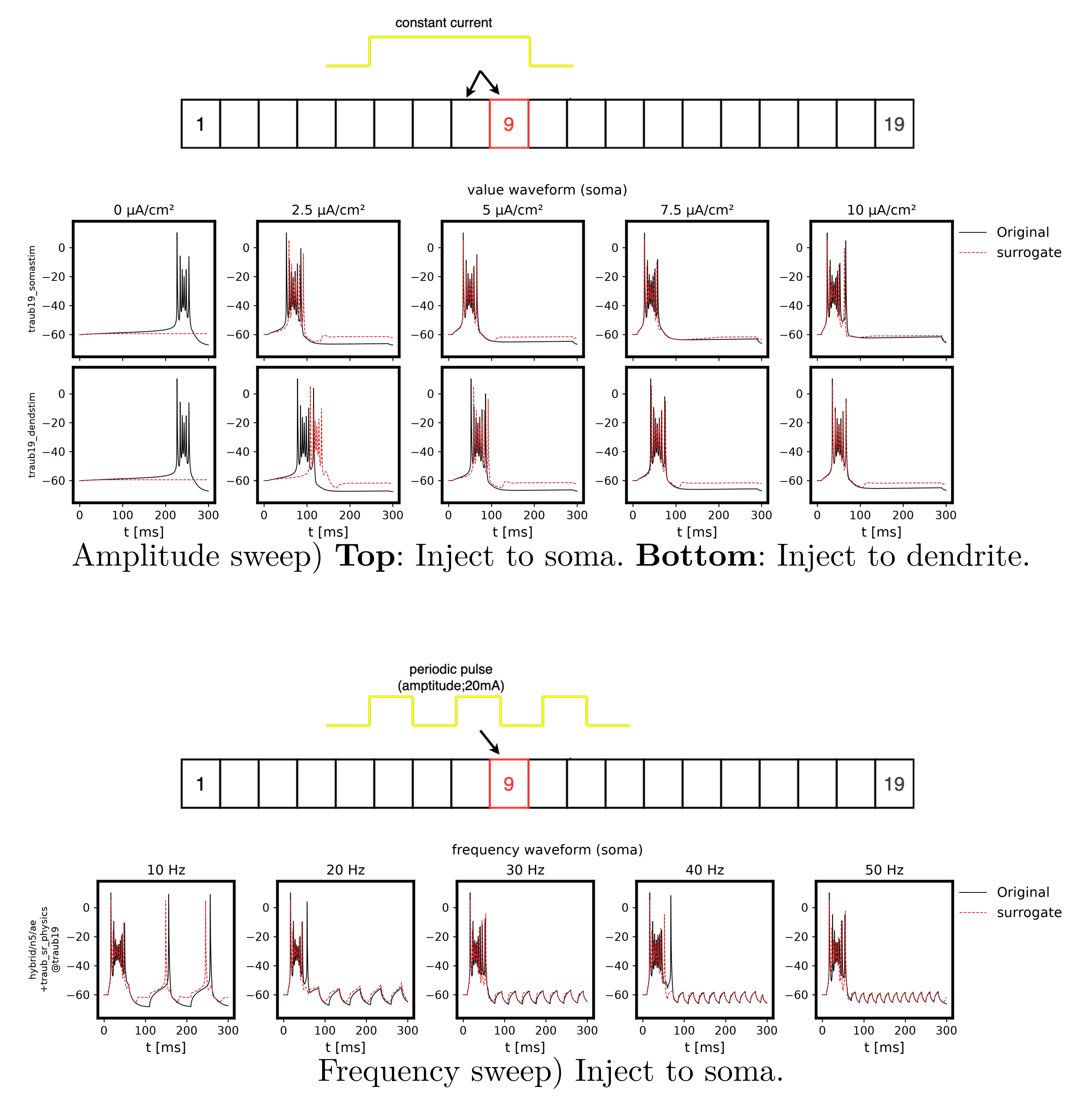
Action Potential reproduction



20 ms, 3 μ A/cm² step: V and the **5 AE latents**.

- latency err **0.3 ms**, AHP timing gap **3.1 ms**.
- peak **13 mV** low (amplitude gap **12.9 mV**), rise / fall rate gap **21.0 / 10.3 mV/ms** — yet AHP depth gap is only **0.28 mV**.

Replace soma compartment to surrogate model



Amplitude sweep) **Top**: Inject to soma. **Bottom**: Inject to dendrite.

Frequency sweep) Inject to soma.

SINDy Coef : 79.6% non-zero — not sparse.

Conclusion

- Latent dynamics preserve **timing**.
- Error concentrates in the **fast, steep transition** — suggests the **AE encoding (n=5)**, not the SINDy library, fails to preserve it.
- Replaced **all 19 compartments**, no divergence; transfers to **unseen site • frequency**.
- **Core finding**: SINDy succeeds **not because we compressed (n: 6→5)** — ⑤ shows the **same result uncompressed (n = 6)**, so it is the **coordinate change itself** that makes gate dynamics learnable.
- Cost reduction follows **only if compression matters** — so it remains **unproven**: ξ is still dense (**79.6%** non-zero).

Code — github.com/MunechikaHaruki/SINDyNeuroSurrogate

[1] [Online]. Available: https://neuromorpho.org/neuron_info.jsp?neuron_name=L16a

[2] R. D. Traub, R. K. Wong, R. Miles, and H. Michelson, “A Model of a CA3 Hippocampal Pyramidal Neuron Incorporating Voltage-Clamp Data on Intrinsic Conductances,” *Journal of Neurophysiology*, vol. 66, no. 2, pp. 635–650, Aug. 1991, doi: 10.1152/jn.1991.66.2.635.

[3] K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, “Data-Driven Discovery of Coordinates and Governing Equations,” *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.

→ **Future:** sparsify ξ , push n down, test on other channel models.

Conflicts of Interest — The authors declare no conflicts of interest regarding this manuscript.